

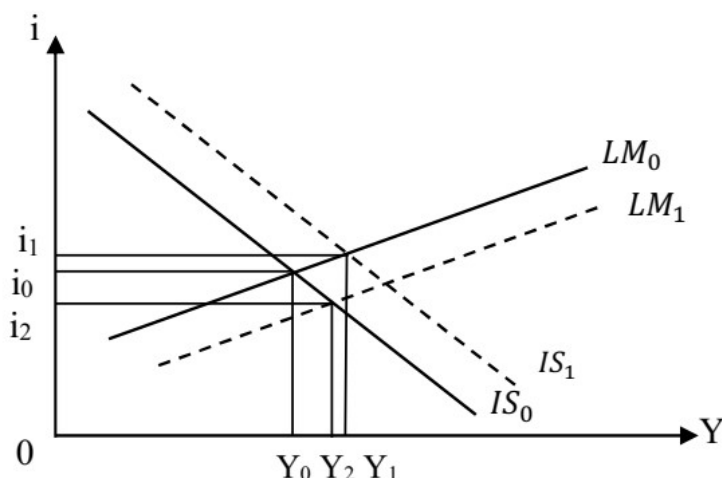


Fig. 3.4: Simultaneous Equilibrium

3.4.1 Shocks in the IS-LM Model

Previously we have discussed about the factors that lead to a shift in the IS curve and the LM curve. Often the economy experiences exogenous demand shocks and supply shocks. Let us analyse the impact of such shocks in terms of the IS-LM model. Suppose there is a boom in the stock market, which changes the wealth of households. Such a situation will increase the level of consumption in the economy. Consequently, there will be an increase in aggregate expenditure and the IS curve will shift to the right from IS_0 to IS_1 (see Fig. 3.4). As a result, there is an increase in output from Y_0 to Y_1 . A negative shock such as a decrease in business confidence of firms will shift the IS curve to the left. Such a left-ward shift in the IS curve will decrease output as well as interest rate.

Let us consider another example. Suppose there is a series of online financial fraud cases in the country. This will send a negative signal to people – this will discourage people to keep their income in the formal system; there will be a decline in online transactions and the demand for money will increase. As a result, the LM curve will shift to the right from LM_0 to LM_1 (see Fig. 3.4). Consequently, there will be an increase in output from Y_0 to Y_2 , and decrease in interest rate from i_0 to i_2 .

You should note that the nature and extent of change in output and interest rate will depend upon the slope of the IS and LM curves. If the LM curve is flatter, monetary policy is quite effective. A small decrease in interest rate by the central bank will increase output by a large amount. If the LM curve is steeper, monetary policy will not be effective. In order to increase output by a small amount we will need a large decrease in the interest rate.

3.4.2 Adjustment Process in the Economy

A question arises at this point: What happens to the economy if it is operating at a point which is not at the intersection of the IS and LM curves? Does it automatically converge to the equilibrium point? Let us try to answer this question.

If the economy is not operating at the unique point where the IS and LM curves intersect, there is disequilibrium in the economy. In that case an adjustment process starts and the economy moves to an equilibrium point. The following two issues are important in this context:

- a) If there is an excess demand for money (to the right of the LM curve), the interest rate will increase. The interest rate will fall in case we observe excess supply in the money market. Fall in interest rate will lead to increase in investment. Increase in investment will affect the IS curve.
- b) Excess demand for goods (to the left of the IS curve) leads to an increase in output. Increase in output will lead to increased demand for money. Increase in demand for money will lead to rise in interest rate and outward shift in the LM curve.

We observe from the above that both real sector and monetary sector are inter-related. A real shock to the economy will have effects on the monetary sector and *vice versa*.

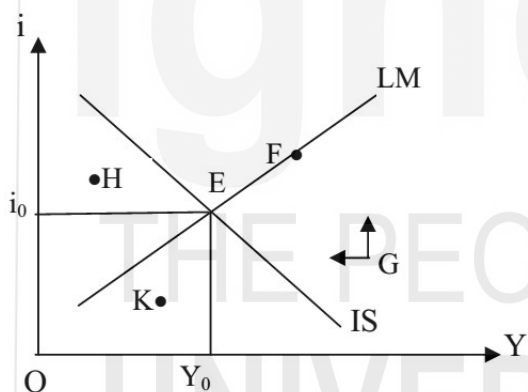


Fig. 3.5: Convergence to Equilibrium Point

In Fig. 3.5 we plot the IS and LM curves. The economy is in equilibrium at point E, with i_0 to Y_0 as equilibrium interest rate and output respectively. Suppose, due to certain shock to the economy, the equilibrium is disturbed. We have indicated the points F, G, H and K where the economy is not in equilibrium.

Let us consider point G. As mentioned earlier, at point G, there is excess demand for money and excess supply of goods. The arrows given at point G specify the direction in which adjustment takes place. Due to excess demand for money, there will be a rise in the rate of interest. This is because people will sell bonds so as to hold more of liquid cash, as a result of which the bond prices will decline and rate of interest will increase. Upward pointing arrow in Fig. 3.5 represents rising interest rate. At point G, there is excess supply of goods also. This will lead to involuntary accumulation of goods; inventories will increase. As a result, firms will cut down production and the level of output will decline. The leftward pointing arrow represents the declining output. Finally, the economy will move towards equilibrium point E.

Points F, H and K in Fig. 3.5 also are disequilibrium points. You can analyse the supply and demand conditions at these points and find out the directions in which adjustment will take place. For example, at point 'H' there is excess supply of money (since it is to the left of the LM curve) and excess demand for goods (since it is to the left of the IS curve). Excess supply of money will lead to a decrease in interest rate. Excess demand for goods will lead to an increase in output level. Consequently, the economy will move towards the equilibrium point E.

Suppose the economy is at point F (see Fig. 3.5) – here the monetary sector is in equilibrium, but the real sector is in disequilibrium. There is an excess supply of goods in the economy. This will lead to an increase in inventory accumulation. As a result, firms will decrease their production levels. Decrease in output will lead to a decrease in the demand for money and consequent decrease in the rate of interest. Finally, the economy will reach the equilibrium point E where both the IS and LM curves intersect.

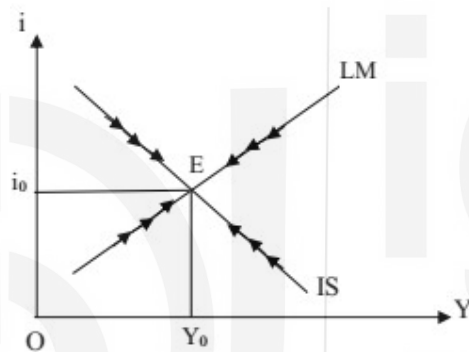


Fig. 3.6: Adjustment Process in the Economy

Note that when one of the two markets is observing disequilibrium, the economy reaches back to equilibrium after the adjustment process. In Fig. 3.6 we have indicated the direction of change by arrow marks.

3.4.3 Impact of Supply Shocks

In the IS-LM model it is not necessary that the equilibrium occurs at the level of potential output of the economy (or, full employment output). Equilibrium could be at any level of output in the economy. An objective of policy makers has always been that the economy should operate at potential output level with unemployment at its natural level. In the long run does the economy maintain equilibrium at the full employment output level?

In the long run prices do not remain constant. If actual output is above the potential out, there will be price rise. The aggregate expenditure of the economy (such as consumption, investment and government expenditure) describes the demand for goods and services, which refer to physical quantities of goods and services. A change in the price level may not affect these quantities. Thus, the IS curve is relatively stable – it may not shift. The LM curve, on the other hand, is based on the supply of and demand for real balances. Thus, a change in price level will shift the LM curve.